

1. (20 points) Consider the public keys (e_1, n_1) and (e_2, n_2) of two RSA cryptosystems.
- a. You have discovered that n_1 and n_2 have a common factor but do not know what it is. How would you find it?

Answer: We know that $n_1 = p * q_1$ and $n_2 = p * q_2$. We also know they share a common factor. Since they are prime, we would simply have to calculate the greatest common factor of the two, which would give us p . Otherwise, the greatest common factor would just be 1.

- b. You have intercepted a message c enciphered using the first public key. You also know the common factor of n_1 and n_2 . Show how to decrypt c .

Answer: Once we find p , you can divide to get the other prime $q_1 = n_1/p$ and then compute the totient $\phi(n_1) = (p - 1)(q_1 - 1)$. We would then want to use this to find the private key. We have the equation $ed \bmod \phi(n) = 1$ and we can rearrange this so that $d_1 = e_1^{-1} \bmod \phi(n_1)$. Lastly, we can use d_1 to decrypt c with the equation $m = c^{d_1} \bmod n_1$ to get the original message (m).